

CRM NO 190643000269	Name B/o. REVATHI 2209259	Father Name KARTHICK	Lab ID 21200106106
Package NBS Trio	Gender MALE	DOB/Age 02/12/2022 (4 Days 1 Hrs)	Referred By Dr. RAMA KRISHNAN
Hospital name SKS Hospital	City SALEM	Pre Term / Full Term Full term	Baby Weight (Kg) 3.000
Specimen Source Venous Blood and Urine	Specimen Notes Acceptable	Collection Date & Time 06/12/2022 06:00	Received Date & Time 10/12/2022 22:15
Date of Report & time 12/12/2022 05:06	Blood Transfusion NO	Transfusion Date NO	Special Feeds / IVF / TPN/ Supplements Yes

INBORN ERRORS OF METABOLISM (IEM) SUMMARY REPORT

Sr. No.	Test Methodology	Result	Test Type	Page
1	FLUOROMETRIC ENZYME IMMUNOASSAY SCREENING REPORT	Negative	Screening	2
2	TANDEM MASS SPECTROMETRY SCREENING REPORT	Negative	Screening	3
3	GAS CHROMATOGRAPHY MASS SPECTROMETRY SCREENING REPORT	Negative	Screening	5

DISCLAIMER

All investigations have their limitations which are imposed by the limits of sensitivity and specificity of individual assay procedures as well as the quality of specimen received by the laboratory.

It is presumed that the received specimen belongs to the patient named or identified in the test request form.

Partial reproduction of this report is not permitted.

It is important to monitor baby's health and take a proactive approach should there be any unhealthy condition(s) within or beyond scope of this test.

A negative screening assay result does not rule out the possibility of an underlying metabolic / genetic disease.

The metabolites and health conditions mapping is not mutually unique and there can be other forms in which an abnormality may be apparent. Similarly, the health condition may not exclusively be an outcome of the disorders in consideration.

If the newborn is premature, retest is suggested when adjusted gestational age is 40 weeks.

HB-Variant hemoglobins may require further testing by the laboratory to diagnose.

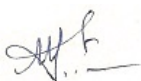
HB-Presence of Hb Bart's with any newborn screen result indicates the presence of alpha thalassemia of unknown severity.

HB-If an increased amount of adult hemoglobin is noted, the infant is presumed to have been transfused (Including Intrauterine transfusion).

HB-Carriers of beta thalassaemia and HbLepore cannot be excluded at this age.

Investigations performed is not intended to replace the medical advice and or treatment obtained from a qualified healthcare practitioner as it doesnot cover all the metabolic disorders

HB (Covered under Hemoglobinopathies package)



Dr. Mahesh Hampe , MD

Lab Director

REPORT- SUMMARY

Observations:

All the analytes are within the normal range.

Interpretation:

The tested sample is screened **NEGATIVE** for the list of disorders tabulated in page 2.

Recommendation:

Please correlate the results with other clinical and diagnostic findings.

Initial Report

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Duplicate Report

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Revision Report

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Version No

1

Disorder	Parameter	Result*	Result Value	Reference	Remarks
					
Glucose 6 Phosphate Dehydrogenase deficiency	G6PD		24.6	NLT 3 U/dL	Normal
Classical Galactosemia	T GAL		2.1	NMT 25 mg/dL	Normal
Congenital Hypothyroidism	TSH**		0.100	NMT 20 µU/mL	Normal
Congenital Adrenal Hyperplasia	17-OHP		4.397	NMT 55 ng/mL	Normal
Profound Biotinidase deficiency	BTD		189.52	NLT 40 U	Normal
Cystic Fibrosis	IRT		6.2531	NMT 110 µg/L	Normal

* This Graph represents the value corresponding to results values. Each box represents the reference value. Results value is shown as dot.

* **NMT** - Not More Than, **NLT** - Not Less Than. ** TSH values are represented in Serum unit (Conversion factor: 1 µU/mL blood = 2.22 µU/mL serum).

G6PD - Glucose-6-Phosphate Dehydrogenase, T GAL - Total Galactose (galactose and galactose-1-phosphate), TSH** - Thyroid Stimulating Hormone, 17-OHP - 17a-OH-progesterone, BTD - Profound Biotinidase, IRT - Human Immunoreactive Trypsin(ongen)

REFERENCES

- <https://www.aap.org/en-us/advocacy-and-policy/aap-health-initiatives/PEHDIC/Pages/Newborn-Screening.aspx>
- Sudhanshu S, Riaz I, Sharma R, Desai MP, Parikh R, Bhatia V. Newborn Screening Guidelines for Congenital Hypothyroidism in India: Recommendations of the Indian Society for Pediatric and Adolescent Endocrinology (ISPAE) – Part II: Imaging, Treatment and Follow-up. The Indian Journal of Pediatrics. 2018 Jun 1;85(6):448–53.

** End Of Report **

REPORT- SUMMARY

Observations:

All the analytes are within the normal range.

Interpretation:

The tested sample is screened NEGATIVE for the list of disorders in Annexure 1.

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Analyte Name	Analyte Result ($\mu\text{mol/L}$)	Analyte Reference	MoM*	MoM Value	MoM Reference
			0.01 0.1 1 10 100		
Alanine	512	71.00-1361.24		2.09	NMT 5.56
Arginine	1.69	0.78-60.00		0.7	NMT 24.69
Citruline	8.93	3.86-62.11		0.78	NMT 5.45
Glutamine	451	48.24-1309.25		0.88	NMT 2.56
Glycine	373	117.35-1270.00		0.97	NMT 3.29
leucine\isoleucine\hydroxyproline	128	23.93-383.93		1.01	NMT 3.02
Methionine	8.11	4.63-48.00		0.49	NMT 2.93
Ornithine	57.4	22.75-330.00		0.66	NMT 3.77
Phenylalanine	54.0	17.10-193.00		1.01	NMT 3.61
Proline	106	30.00-448.00		0.77	NMT 3.27
Tyrosine	56.3	15.34-455.88		0.75	NMT 6.10
Valine	113	31.28-450.00		0.99	NMT 3.95
Free Carnitine	28.1	7.00-121.04		1.12	NMT 4.82
Acetylcarnitine	29.8	2.49-65.00		1.81	NMT 3.81
Propionylcarnitine	3.50	0.12-7.02		2.15	NMT 4.31
Butyrylcarnitine	0.30	0.06-1.30		1.36	NMT 6.27
Isovalerylcarnitine	0.13	0.00-0.60		1.08	NMT 5.00
Tiglylcarnitine	0.01	0.00-0.14		1	NMT 14.00
Hexanoylcarnitine	0.04	0.00-0.30		1	NMT 7.50
Octanoylcarnitine	0.03	0.00-0.37		0.5	NMT 6.17
Octenoylcarnitine	0.04	0.00-0.32		1.33	NMT 10.67
Decenoylcarnitine	0.02	0.01-0.24		0.5	NMT 6.00
Decadienoylcarnitine	0.01	0.00-0.07		1	NMT 7.00
Tetradecenoylcarnitine	0.06	0.01-0.51		0.86	NMT 7.30
Hexadecanoylcarnitine	4.24	0.00-9.04		1.18	NMT 2.53
3-Hydroxy-hexadecanoylcarnitine	0.01	0.00-0.09		0.5	NMT 4.50
Octadecanoylcarnitine	1.11	0.00-2.41		1.19	NMT 2.59
Octadecenoylcarnitine	1.66	0.00-3.25		1.33	NMT 2.60
Succinylacetone	0.26	0.00-1.00		1.04	NMT 4.00
Adenosine	0.79	0.00-3.52		1.34	NMT 5.97
2-deoxyadenosine	0.04	0.00-0.10		2	NMT 5.00
C26:0-lysophosphatidylcholine	0.14	0.0 - 1.06		0.47	NMT 3.53
Argininosuccinic acid	0.31	0.00-2.02		1.35	NMT 8.70
Glutamic acid	532	47.00-915.00		2.12	NMT 3.65
Propionylcarnitine/Acetylcarnitine	0.12	0.00-0.55		1.09	NMT 5.00
Methylmalonyl	0.28	0.00-1.50		1.65	NMT 8.82
Glutaryl carnitine	0.09	0.00-0.35		1.29	NMT 5.00
Malonylcarnitine	0.15	0.00-1.00		1.5	NMT 10.00

* This Graph represents the value corresponding to Multiple of Median(MoM). Each box represents the interval between 1%ile and 99%ile. The MoM Shown as dot.

ANNEXURE - 1 LIST OF DISORDERS

Amino Acid Disorders	Organic Acid Disorders
Citrulinemia Type I	Long-chain L-3 hydroxyacyl-CoA dehydrogenase deficiency
Citrulinemia Type II	Malonic acidemia
Hyperornithinemia with Gyral Atrophy	Methylmalonic acidemia (Cobalmin Disorders)
5-Oxoprolinuria	Methylmalonic acidemia (Mutase)
Transient Neonatal Tyrosinemia	3-Methylcrotonyl-CoA carboxylase deficiency
Argininosuccinic Acidemia	Multiple Carboxylase Deficiency (also known as Holocarboxylase synthase deficiency)
Argininemia	3-Methylglutaconic aciduria
Defects of bipterin cofactor biosynthesis	2-Methyl-Butyryl-Glycinuria
Disorders of bipterin cofactor regeneration	2-Methyl-3-HydroxyButyric Aciduria
Benign Hyperphenylalaninemia	Propionic acidemia
Homocystinuria	Other Conditions
Hypermethioninemia	Adenosine deaminase (ADA)–severe combined
Maple syrup (urine) disease	Carbamoyl phosphate synthetase I
Non-ketotic hyperglycinemia	Hyperornithinemia-Hyperammonemia-Homocitrullinuria
Phenylketonuria	Hyperalimentation (TPN)
Tyrosinemia type I	Liver Complications
Tyrosinemia type II	Medium Chain Triglyceride (MCT) Oil Administration
Tyrosinemia type III	N-acetylglutamate synthase deficiency
Fatty Acid Oxidation Disorders	Ornithine transcarbamylase deficiency
Carnitine Palmitoyl Transferase Type I Deficiency	Treatment with antibiotics/Pyvalic Acid
Carnitine Uptake Defect	Treatment with Benzoate or Valproic Acid
Carnitine Acyl-Carnitine Translocase deficiency	X-linked adrenoleukodystrophy
Carnitine Palmitoyl-Transferase type II deficiency	
2,4 Dienoyl-CoA reductase deficiency	
Glutaric acidemia type II	
Medium-Chain Acyl-CoA Dehydrogenase Deficiency	
Medium-chain ketoacyl-CoA thiolase deficiency	
Medium/short-chain L-3-hydroxyacyl-CoA dehydrogenase deficiency	
Short-Chain Acyl-CoA Dehydrogenase Deficiency	
Trifunctional protein deficiency	
Very long-chain acyl-CoA dehydrogenase deficiency	
Organic Acid Disorders	
Isobutyryl-CoA-Dehydrogenase Deficiency	
Glutaric Acidemia type I	
3-Hydroxy-3-Methyl-Glutaric aciduria	
Iso-Valeric Acidemia	
Beta-KetoThiolase deficiency	

REFERENCES

1. Ombrone D, Giocaliere E, Forni G, Malvagia S, la Marca G. Expanded newborn screening by mass spectrometry: New tests, future perspectives. Mass Spectrom Rev. 2016 Feb;35(1):71–84.
2. Gaugler S, Rykl J, Wegner I, Von Däniken T, Fingerhut R, Schlotterbeck G. Extended and Fully Automated Newborn Screening Method for Mass Spectrometry Detection. International Journal of Neonatal Screening. 2018 Mar;4(1):2.

** End Of Report **

REPORT- SUMMARY

Observations:

All the analytes are within the normal range.

Interpretation:

The tested sample is screened **NEGATIVE** for the list of disorders in Annexure 1.

Recommendation:

Please correlate the results with other clinical and diagnostic findings.

Initial Report

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Analyte Name	Analyte Result (mmol/mol of creatinine)	Analyte Reference	Log of MoM*	MoM Value	MoM Reference
			-5 -4 -3 -2 -1 0 1 2 3 4 5		
2-Deoxytetronic acid-3	181.69	NMT 806.88		0.27	NMT 0.91
2-Hydroxyglutaric acid-3	49.98	NMT 433.03		0.02	NMT 0.96
3-Hydroxy-3-methylglutaric acid-3	18.58	NMT 152.82		0.14	NMT 1.05
3-Hydroxyadipic acid-3	19.51	NMT 597.47		-0.32	NMT 1.17
3-Hydroxybutyric acid-2	236.52	NMT 25913.94		0.46	NMT 2.50
3-Hydroxyisovaleric acid-2	29.86	NMT 330.38		0.45	NMT 1.49
3-Hydroxypropionic acid-2	53.41	NMT 473.32		0.5	NMT 1.45
4-Hydroxyphenylacetic acid-2	81.79	NMT 773.55		0.41	NMT 1.39
4-Hydroxyphenylpyruvic acid-3	71.08	NMT 443.93		1.23	NMT 2.03
Adipic acid-2	0.49	NMT 296.01		-1.4	NMT 1.42
Ethylhydracrylic acid-2	2.92	NMT 138.32		-0.46	NMT 1.22
Galactitol-6	69.83	NMT 788.22		0.26	NMT 1.31
Galactose-5	306.59	NMT 4053.36		0.24	NMT 1.36
Glutaconic acid-3	3.29	NMT 9.34		1.16	NMT 1.61
Glyceric acid-3	37.94	NMT 424.53		0.31	NMT 1.36
Glycerol 3-phosphate-4	32.33	NMT 203.13		0.33	NMT 1.13
Glycerol-3	436.91	NMT 19571.15		0.39	NMT 2.04
Glycine-3	105.08	NMT 1722.91		0.21	NMT 1.42
Glycolic acid-2	192.99	NMT 1238.59		0.41	NMT 1.22
Homocysteine-3	11.63	NMT 77.89		0.14	NMT 0.96
Homovanillic acid-2	5.38	NMT 316.91		-0.21	NMT 1.57
Lactic acid-2	551.98	NMT 21683.55		0.37	NMT 1.96
Malic acid-3	16.29	NMT 413.27		0.2	NMT 1.61
Methylcitric acid-4	30.48	NMT 2014.08		-0.23	NMT 1.59
Methylcitric acid-4	1.9	NMT 24.34		0.29	NMT 1.40
Methylmalonic acid-2	23.89	NMT 1216.56		0.4	NMT 2.10
Vanillylmandelic acid-3	3.7	NMT 47.81		-0.06	NMT 1.05

* This Graph represents the metabolites which are found above the detection limit. Each box represents the reference interval between 1%ile and 99%ile Multiple of Median(MoM) in Logarithmic scale. The Log of MoM of the analyzed sample is shown as dot.

ANNEXURE - GCMS LIST OF DISORDERS

Amino Acidopathies and Organic Acidemias	Amino Acidopathies and Organic Acidemias	TCA Cycle/Mitochondrial Abnormality
2-Hydroxyglutaric aciduria	Hyperleucine-isoleucinemia	Pyruvate dehydrogenase phosphatase deficiency
2-ketoadipic aciduria	Hyperglycinuria (Ketotic)	Disorders of Fatty Acid Metabolism
3-hydroxy-3-methylglutaryl-CoA-lyase deficiency	Hyperglycinuria (non-ketotic)	Dicarboxylic aminoaciduria
3-Hydroxyisobutyryl CoA Deacylase Deficiency	Hypermethioninemia	Ethyl Malonic Aciduria
3-methylcrotonyl CoA carboxylase deficiency	Hypersarcosinemia	Glycerol Kinase Deficiency
3-methylglutaconic aciduria Type I	Hyperlysinemia Type 1	Medium chain acyl CoA dehydrogenase deficiency
5-oxoprolinuria (also known as pyroglutamic aciduria)	Isobutyryl-CoA dehydrogenase deficiency (IBD)	Mitochondrial trifunctional protein deficiency
Alkaptonuria	Iminoglycinuria	Very Long-chain acyl-CoA dehydrogenase deficiency
Aminoacylase I Deficiency	Imidazole Amino Aciduria	Short chain acyl CoA dehydrogenase dehydrogenase deficiency
Argininemia	Isovaleric acidemia	Zellweger like syndrome
Argininosuccinic aciduria	Lysinuric protein intolerance	Peroxisomal Disorders
Benign hyperphenylalaninemia	Malonic acidemia	Infantile refsum disease
Beta-Aminoisobutyric Aciduria	Maple syrup urine disease (MSUD)	Neonatal Adrenoleukodystrophy
Profound Biotinidase deficiency	Methylmalonic aciduria, cblA and cblB forms	Primary hyperoxaluria Type 1
Beta- ketothiolase deficiency (BKT)	Methylmalonic acidemia and Homocystinuria -Cbl C, D	Primary hyperoxaluria Type 2
Canavan disease	Methylmalonic semialdehyde dehydrogenase deficiency	Zellweger syndrome
Carbamoyl Phosphate Synthetase-1 Deficiency / N-Acetylglutamate Synthetase Deficiency	Methylmalonyl- CoA mutase deficiency (MUT)	Disorders of Purine and Pyrimidine Metabolism
Carnosinuria	Mevalonic aciduria	Adenine phosphoribosyl transferase deficiency
Citrullinemia Type 1	Multiple carboxylase deficiency	Dihydropyrimidinase Deficiency
Citrullinemia type II	N-Acetylglutamate Synthetase Deficiency (merged with CPS1)	Hyperuric acidemia
Cystinuria	NICCD	Lesch-Nyhan syndrome
Cystathioninuria	Ornithine Trans Carbamylase Deficiency	Molybdenum cofactor deficiency
Defects of bioppterin cofactor regeneration (BIOPT REG)	Phenylketonuria (PKU)	Orotic aciduria
Dihydropolipoyl Dehydrogenase (E3) Deficiency (also known as MSUD Type 3)	Propionic acidemia	Partial deficiency of hypoxanthine-guanine phosphoribosyltransferase
Familial Renal iminoglycinuria	Saccharopinuria (also known as Hyperlysinemia Type II)	Thymine Uraciluria
Formiminoglutamic aciduria	Succinic semialdehyde dehydrogenase deficiency	Xanthinuria (Type I & II)
Glutaric aciduria type II	Tyrosinemia caused by liver dysfunction	Disorders of Sugars
Glutaric aciduria type I	Transient neonatal tyrosinemia	D-glyceric aciduria
Glutathionuria	Tryptophanuria with dwarfism	Endogenous sucrosuria
Hartnup Disease	Tyrosinemia Type II	Fructose-1 and 6-diphosphatase deficiency
Hawkinsinuria	Tyrosinemia Type III (also covers transient neonatal tyrosinemia)	Fructosuria
Hyperornithinemia-hyperammoninemia-hyperhomocitrullinemia (HHH) syndrome	Tyrosinemia Type I	Galactokinase deficiency (GALK)
Histidinuria	Valinemia	Galactose epimerase deficiency (GALE)
Histidinemia	Xanthurenic aciduria	Lactic Acidosis
Homocystinuria	TCA Cycle/Mitochondrial Abnormality	Transient Galactosemia
Hyperprolinemia type I	Fumarate hydratase deficiency	Non-IEM Disorder
Hyperprolinemia type II	Leigh syndrome	Neuroblastoma
Hydroxylysinuria	Pyruvate carboxylase deficiency	
Hyperhydroxyprolinemia	Pyruvate decarboxylase deficiency	
	Pyruvate dehydrogenase (E1) deficiency	

REFERENCES

1. Ombrone D, Gicaliere E, Forni G, Malvagia S, la Marca G. Expanded newborn screening by mass spectrometry: New tests, future perspectives. Mass Spectrom Rev. 2016 Feb;35(1):71–84.
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** End Of Report **